Chapter 4
Measuring Systems



4.1 Measurement: Forms of Realization

Measurements can be carried out in two ways: direct and indirect. In this section, we will address these two modalities and their particularities.

4.1.1 Direct Measurement

Direct measurement occurs when only a quantity is involved in the process, and the instrument is used directly to obtain the desired measurement result.

Some examples of direct measurements:

- Diameter measurement of a cylinder with a Vernier caliper
- Weighing an object with a scale
- Measurement of the electric current of a circuit with an ammeter
- Pressure indication using a Bourdon-type pressure gauge.

4.1.2 Indirect Measurement

It occurs when measurements involve one or more related quantities through a mathematical equation.

Examples:

- Determination of the area (A) of a rectangular terrain measuring the length of each of its sides L1 and L2. We adopted the expression: $A = L1 \times L2$.

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P. P. Novellino do Rosario, A. Mendes, *Metrology and Measurement Uncertainty*,
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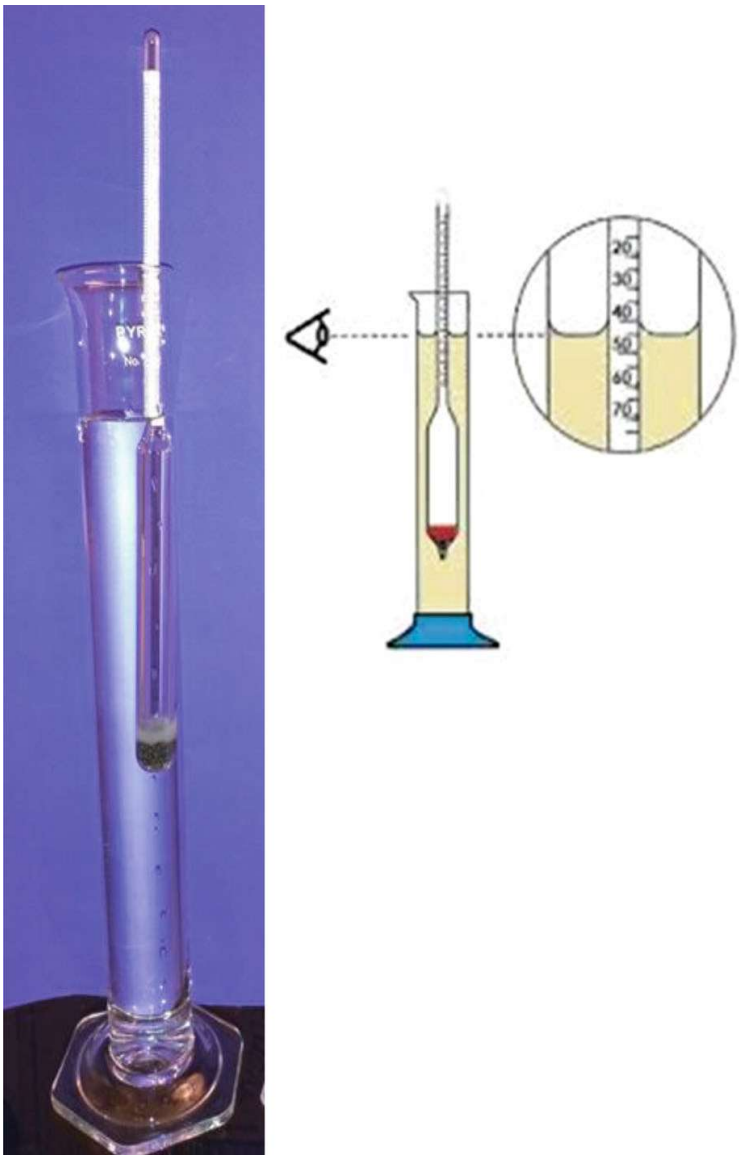


Fig. 4.2 Float densimeter. (https://http2.mlstatic.com)

4.2 Metrological Characteristics of Measurement Systems

Measurement systems have several metrological characteristics described in the International Vocabulary of Metrology. In this section, we will highlight the most usual.

4.2.1 Indication Interval

According to VIM—4.3, the indication interval is:

Set of quantity values bounded by extreme possible indications.

NOTE 1 An indication interval is usually stated in terms of its smallest and greatest quantity values, for example, "99 V to 201 V."

NOTE 2 In some fields, the term is "range of indications"



Fig. 4.5 Digital multimeter. (https://pixabay.com/vectors/device-electric-electronics-measure-1296017/)

4.2.2 Measuring Interval (Working Interval)

The VIM—4.7 is defined as:

> *Set of values of quantities of the same kind that can be measured by a given measuring instrument or measuring system with specified instrumental measurement uncertainty under defined conditions.*
>
> *NOTE 1 In some fields, the term is "measuring range" or "measurement range"*
>
> *NOTE 2 The lower limit of a measuring interval should not be confused with the detection limit*

The measurement interval is lower or, at most, equal to the indication interval and can be obtained in manuals, technical standards, or calibration reports.

Example 4.1
The 3 ½ digit digital multimeter (Fig. 4.5) measures continuous electrical voltage with an indication interval of (0 to 1000) V. However, this interval is subdivided into the following measurement intervals: (0 to 200) mV; (0 to 2000) mV; (0 to 20) V; (0 to 200) V; and (0 to 1000) V.



Fig. 4.6 Manometer: vacuum and positive pressure. (https://pixabay.com/photos/pressure-meter-engineering-gauge-2113401/)

4.2.3 Range of a Nominal Indication Interval

The VIM—4.5 defines the range as follows:

> *Absolute value of the difference between the extreme quantity values of a nominal indication interval.*
>
> *EXAMPLE: For a nominal indication interval of −10 V to +10 V, the range of the nominal indication interval is 20 V.*
>
> *NOTE Range of a nominal indication interval is sometimes termed "span of a nominal interval."*

Example 4.2
Note that the working interval of the gauge shown in Fig. 4.6 is (−100 to 500) kPa, but its measurement range is:

$$\text{Range} = [500 - (-100)]\ \text{kPa} = 600\ \text{kPa}$$

4.2.4 Division of Scale (Not in VIM)

It is the difference between the scale values corresponding to two successive marks. The unit scheduled on the scale expresses the value of a division, whatever the unit of the measurement (Fig. 4.7).

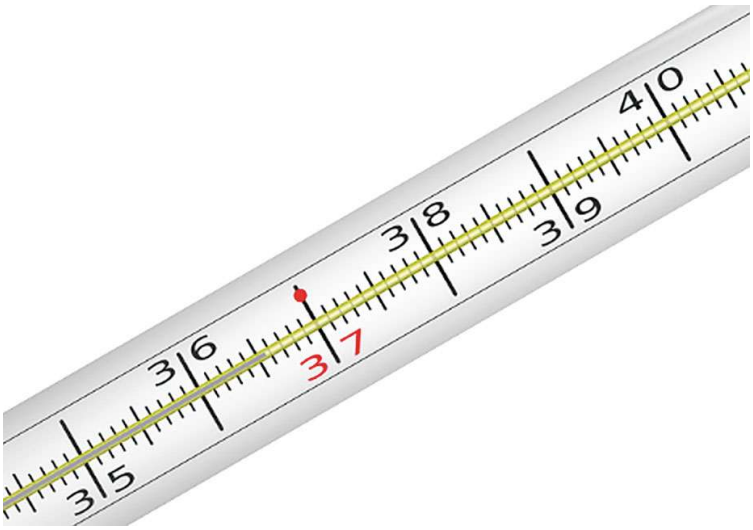


Fig. 4.7 Clinical thermometer with division of scale equal to 0.1 °C. (https://pixabay.com/vectors/clinical-thermometer-fever-153666)



Fig. 4.8 Manometer with division of scale equal to 10 kPa and resolution of 5 kPa. (https://pixabay.com/photos/pressure-meter-engineering-gauge-2113401/)

4.2.5 Resolution of a Displaying Device

The VIM—4.15 defines it as the *"smallest difference between displayed indications that can be meaningfully distinguished."*

The operator should evaluate the reading resolution in systems with analog dials (Fig. 4.8).



Fig. 4.11 Pierre Vernier (1580–1637). (https://alchetron.com/cdn/pierre-vernier-7b642191-a050-4c17-8bd2-7dea18d3bb1-resize-750.jpeg)

Knowing a Little More... (Fig. 4.10) (continued)
describes his invention of a senior board and a method to determine the angles of a triangle with its known sides.

Source: Adapted from https://www.biografias.es/famosos/pierre-vernier.html.

4.2.6 Sensitivity of a Measuring System

According to VIM—4.12, the sensitivity of a measuring system is:

> *Quotient of the change in an indication of a measuring system and the corresponding change in a value of a quantity being measured.*
>
> *NOTE 1 Sensitivity of a measuring system can depend on the value of the quantity being measured*
>
> *NOTE 2 The change considered in the value of a quantity being measured must be large compared with the resolution*

Example 4.4

(a) A Pt-100 type platinum resistance thermometer has a sensitivity of 0.38 Ω/°C, that is, each one °C stimulus in temperature causes a variation in the electrical resistance of the Pt-100 of 0.38 ohm.

(b) An electrode's sensitivity for pH measurement shall be 59.16 mV/pH, that is, one pH variation in the substance should generate 59.16 mV of electrode output variation.

(c) A type K thermocouple must have a sensitivity of 39.5 mV/°C, and a type J thermocouple must have a 50.4 mV/°C.

4.2.7 Stability of a Measuring Instrument

According to VIM—4.19, it is:

Property of a measuring instrument, whereby its metrological properties remain constant in time

NOTE: Stability may be quantified in several ways.

EXAMPLE 1 In terms of the duration of a time interval over which a metrological property changes by a stated amount

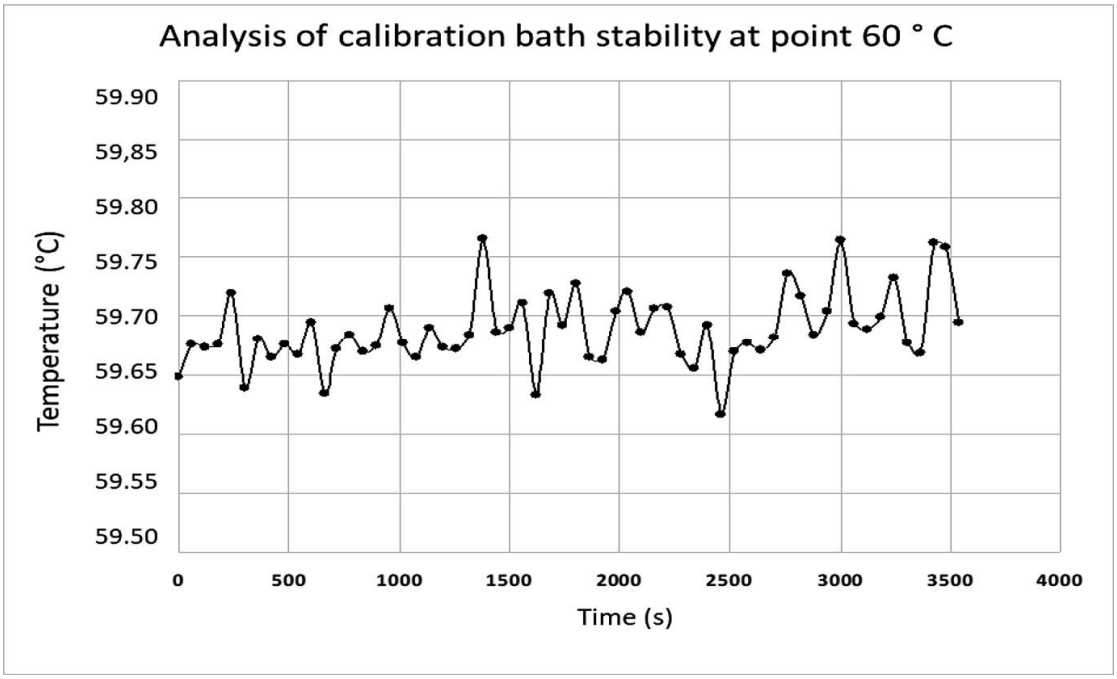
EXAMPLE 2 In terms of the change of a property over a stated time interval

In Table 4.1, we present an example of the analysis of the stability of the temperature of a liquid calibration bath at a point close to 60 °C. This bath has a measurement range between 50 °C and 300 °C.

The stability of a bath is the variation of its temperature at a given point after the bath goes into thermal equilibrium. Since no instrument is stable, this investigation will verify how much the calibration bath temperature oscillates after it is fixed at a given value. In this example, the desired value is 60 °C.

Table 4.1 Calibration bath temperature values at point 60 °C

Time (s)	Temperature (°C)	Time (s)	Temperature (°C)	Time (s)	Temperature (°C)
0	59.65	1201	59.67	2402	59.69
60	59.68	1261	59.67	2462	*59.62*
120	59.67	1321	59.68	2522	59.67
180	59.68	1381	*59.77*	2582	59.68
240	59.72	1441	59.69	2642	59.67
300	59.64	1501	59.69	2702	59.68
360	59.68	1561	59.71	2762	59.74
420	59.67	1621	59.63	2822	59.72
480	59.68	1681	59.72	2882	59.68
540	59.67	1741	59.69	2942	59.70
600	59.69	1801	59.73	3002	59.76
661	59.63	1861	59.66	3062	59.69
721	59.67	1921	59.66	3122	59.69
781	59.68	1981	59.70	3182	59.70
841	59.67	2042	59.72	3242	59.73
901	59.67	2102	59.69	3303	59.68
961	59.71	2162	59.71	3363	59.67
1021	59.68	2222	59.71	3423	59.76
1081	59.67	2282	59.67	3483	59.76
1141	59.69	2342	59.66	3543	59.69



Graph 4.1 Analysis of calibration bath stability at point 60 °C

For this analysis, we collected the bath temperature values for one hour at time intervals of one minute. We use a resistance thermometer (Pt-100 four wires) connected to a 6 ½ digit multimeter.

Note that after one hour of analysis, the highest temperature value is 59.77 °C, and the smallest is 59.62 °C. We can conclude that the stability (E) of the calibration bath, after the thermal equilibrium at point 60 °C, was:

$$E = (59.77–59.62) \ \ ^\circ\text{C} = 0.15 \ \ ^\circ\text{C}$$

Graph 4.1 demonstrates the temperature variation of the calibration bath when it is stable at around 60 °C.

4.3 Errors in Measurement Systems

When we calibrate a measurement instrument, we set a comparison between the values obtained by the calibration instrument and the values provided by the standard. Some metrological characteristics obtained in this comparison are the errors and trends of the instruments. In this section, we will address these definitions and their applications.

4.3.1 Measurement Error

The VIM—2.16 defines measurement error as follows:

Measured quantity value minus a reference quantity value.
NOTE 1 The concept of 'measurement error' can be used in both
(a) *when there is a single reference quantity value to refer to, which occurs if a calibration is made by means of a measurement standard with a measured quantity value having a negligible measurement uncertainty or if a conventional quantity value is given, in which case the measurement error is known, and,*
(b) *if a measurand is supposed to be represented by a unique true quantity value or a set of true quantity values of negligible range, in which case the measurement error is not known.*

NOTE 2 Measurement error should not be confused with production error or mistake

Then, the error is:

$$E = X - Rv \tag{4.1}$$

where E = measurement error; X = measured value; and Rv = reference value.

Usually, the reference value is attributed to the value of the standard.

Mathematically, the measurement error can be positive or negative. A positive error denotes that the instrument measurement is greater than the reference value, and a negative error denotes that the measurement is less than the reference value.

Important
When we do more than one measurement at the same point and get different values for error, we adopt the largest of these values as the measurement error.

Solved Exercise 4.1
Four voltage measurements were performed using a voltmeter. The values found were 127.5 V, 127.6 V, 127.5 V, and 127.4 V. Knowing that the reference value is 127.68 V, determine the voltmeter measurement error.

Solution
The measurement error is given by Eq. (4.1). Therefore, we will have:

$$E1 = (127.5 - 127.68)\ \text{V} = -0.18\ \text{V}$$

$$E2 = (127.6 - 127.68)\ \text{V} = -0.08\ \text{V}$$

$$E3 = (127.5 - 127.68)\ \text{V} = -0.18\ \text{V}$$

$$E4 = (127.4 - 127.68)\ \text{V} = -0.28\ \text{V}$$

Since the voltmeter has four error values, we will adopt the value of the most significant measurement error (in absolute terms).

$$E = -\ 0.28\ \text{V} = -\ 0.3\ \text{V}$$

Important
The measurement error result will be −0.3 V, since we must round it to the same number of decimal digits as the instrument reading from which we are determining the measurement error.

4.3.2 Instrumental Bias and Correction

4.3.2.1 Instrumental Bias

The definition of VIM—4.20 for instrumental bias is *"average of replicate indications minus a reference quantity value."*

We should not confuse instrumental bias with measurement error. The instrumental bias determines the instrument's average measurement error.

$$B = \overline{X} - R_v \tag{4.2}$$

B = instrumental bias; $-X$ = mean of measurements; and R_v = reference value.

Solved Exercise 4.2
Determining the instrumental bias of Solved Exercise 4.1, we have (Table 4.2):

$$B = (127.5 - 127.68)\ \text{V} = -0.18\ \text{V}$$
$$B = -0.2\ \text{V}$$

Important
The result of the bias will be −0.2 V, since we must round the result to the same number of decimal digits as the instrument reading from which we are determining the instrumental bias.

Table 4.2 Measurements and mean

Measurements (V)	Mean (V)	Reference value (V)
127.5	127.5	127.68
127.6		
127.5		
127.4		

4.3.2.2 Correction

According to VIM—2.53, we have the following definition for correction:

> *Compensation for an estimated systematic effect.*
> *NOTE 1 See GUM:1995, 3.2.3, for an explanation of 'systematic effect'*
> *NOTE 2 The compensation can take different forms, such as an addend or a factor, or can be deduced from a table*

Correction is equal to the bias with a changed signal and must be added to the value of indications to compensate for the systematic effect.

In Solved Exercise 4.2, the correction would be +0.2 V, and the value of the corrected voltmeter measurement would be (127.5 + 0.2) V = 127.7 V.

4.3.3 Instrumental Drift

The VIM—4.21 defines instrumental drift as:

> *Continuous or incremental change over time in indication due to changes in metrological properties of a measuring instrument.*
> *NOTE: Instrumental drift is related neither to a change in a quantity being measured nor to a change of any recognized influence quantity.*

It is widespread for an instrument of measurement to vary its metrological properties, such as measurement uncertainty and measurement error, over time. For this reason, we must verify the periodicity of these variations and perform calibrations in the measuring instruments at smaller intervals than their instrumental drift.

To verify the stability of a measurement instrument, we analyze your calibration certificate over two or more consecutive calibrations. We keep the calibration certificates from one period to another (usually from year to year) and compare their uncertainties, trends, and measurement errors.

Solved Exercise 4.3
An analytical scale, class I, with a resolution of 0.1 mg, was calibrated, and the table data for its calibration certificate were obtained (Table 4.3).

Table 4.3 Calibration results

Calibration results					
Indication (g)	Standard (g)	Object (g)	Bias (mg)	Uncertainty (mg)	k
20	20.000011	20.0000	0.0	0.2	2.01
40	40.000028	40.0000	0.0	0.3	2.00
70	70.000021	70.0003	0.3	0.3	2.02
100	100.000010	100.0001	0.1	0.3	2.01
120	120.000021	120.0001	0.1	0.4	2.01
150	150.000020	150.0001	0.1	0.4	2.00
220	220.000041	220.0003	0.3	0.5	2.00

Table 4.4 Data from the certificate

Indication (g)	Standard (g)	Object (g)	Bias (mg)	Uncertainty (mg)	k
20	20.000005	20.0000	0.0	0.2	2.00
40	40.000022	40.0000	0.0	0.4	2.01
70	70.000051	70.0005	0.4	0.4	2.01
100	100.000006	100.0001	0.1	0.4	2.00
120	120.000018	120.0001	0.1	0.5	2.02
150	150.000014	150.0001	0.1	0.5	2.02
220	220.000011	220.0004	0.4	0.5	2.02

Table 4.5 Instrumental Drift

Indication (g)	Bias (mg) (year 1)	Bias (mg) (year 2)	Drift (mg)
20	0.0	0.0	0.0
40	0.0	0.0	0.0
70	0.3	0.4	0.1
100	0.1	0.1	0.0
120	0.1	0.1	0.0
150	0.1	0.1	0.0
220	0.3	0.4	0.1

A year later, it was calibrated again. Table 4.4 shows data from the certificate. Determine the instrumental drift from the scale one year to the next.

Solution

We must subtract the trend values between two consecutive years to determine the instrumental drift from the balance from one year to another. See Table 4.5.

4.3.4 Maximum Permissible Measurement Error

The VIM—4.26 definition is:

Extreme value of measurement error, with respect to a known reference quantity value, permitted by specifications or regulations for a given measurement, measuring instrument, or measuring system.

NOTE 1 Usually, the term "maximum permissible errors" or "limits of error" is used where there are two extreme values

NOTE 2 The term "tolerance" should not be used to designate 'maximum permissible error'

Example 4.5
The standard CEN EN 837-1 Pressure Gauges—Part 1: Bourdon Tube Pressure Gauges—Dimensions, Metrology, Requirements, and Testing defines the following maximum permissible errors for analog manometer concerning its measurement range, such as:

- Class 0.1—maximum error of 0.1 %.
- Class 0.25—maximum error of 0.25 %.
- Class 0.6—maximum error of 0.6 %.
- Class 1—maximum error of 1.0 %.
- Class 1.6—maximum error of 1.6 %.
- Class 2.5—maximum error of 2.5 %.

4.3.5 Hysteresis (Not in VIM)

Hysteresis (H) is the most significant difference, in absolute value, of the charge (C) values (measurement made when applying an increasing signal in value) and (D) discharge (measurement made when applying a decreasing signal in value) of a measurement instrument.

$$H = |C \quad - D| \tag{4.4}$$

Hysteresis is a typical phenomenon in mechanical instruments, with a source of error, especially clearances and deformations associated with friction. Examples of instruments that may present hysteresis errors are scales, dynamometers, and analog gauges.

Solved Exercise 4.4
When calibrating a pressure gauge, we determine its hysteresis by charging (increasing pressure) and discharging (decreasing pressure). Table 4.6 shows the result of one calibration cycle. Determine the gauge's hysteresis at each point.

Solution
Table 4.7 shows how we can determine hysteresis at each point from the previous data by subtracting the charge and discharge values (in absolute value).

The pressure gauge hysteresis will be at its highest value: 0.2 bar.

Table 4.6 Result of one calibration cycle

Value read in gauge (bar)	Charge read in standard (bar)	Discharge read in standard (bar)
10	9.9	10.0
20	19.9	20.1
30	30.0	30.0
40	40.2	40.1
50	50.3	50.1

Table 4.7 Hysteresis

Value read in gauge (bar)	Charge read in standard (bar)	Discharge read in standard (bar)	Hysteresis (bar)
10	9.9	10.0	0.1
20	19.9	20.1	0.2
30	30.0	30.0	0.0
40	40.2	40.1	0.1
50	50.3	50.1	0.2

4.3.6 Measurement Accuracy and Precision

These concepts may be the metrological characteristics with the most application mistakes. It is common for people to change the definition of accuracy with that of precision. It became common sense to call a precise measurement when referring to an exact measurement.

This section will define these terms and check their correct application.

4.3.6.1 Measurement Accuracy

The VIM—2.13 presents the following definition of measurement accuracy:

> *Closeness of agreement between a measured quantity value and a true quantity value of a measurand.*
>
> *NOTE 1 The concept 'measurement accuracy' is not a quantity and is not given a numerical quantity value. A measurement is said to be more accurate when it offers a smaller measurement error*
>
> *NOTE 2 The term "measurement accuracy" should not be used for measurement trueness and the term "measurement precision" should not be used for 'measurement accuracy', which, however, is related to both these concepts*
>
> *NOTE 3 'Measurement accuracy' is sometimes understood as the closeness of agreement between measured quantity values that are being attributed to the measurand.*

The true value would be obtained by a perfect measurement (which does not exist), being, by nature, indeterminate. Since the true value is indeterminate, it is used the *conventional quantity value* [VIM—2.12]:

> *Quantity value attributed by agreement to a quantity for a given purpose.*
>
> *EXAMPLE 1 Standard acceleration of free fall (formerly called "standard acceleration due to gravity"),* $g_n = 9.806\ 65\ m{\cdot}s^2$
>
> *EXAMPLE 2 Conventional quantity value of the Josephson constant,* $K_{J\text{-}90} = 483{,}597.9\ GHz\ V^{-1}$
>
> *EXAMPLE 3 Conventional quantity value of a given mass standard,* $m = 100.003\ 47\ g$
>
> *NOTE 1 The term "conventional true quantity value" is sometimes used for this concept, but its use is discouraged*
>
> *NOTE 2 Sometimes a conventional quantity value is an estimate of a true quantity value*

NOTE 3 A conventional quantity value is generally accepted as being associated with a suitably small measurement uncertainty, which might be zero

Or the *reference quantity value* [VIM—5.18]:

Quantity value used as a basis for comparison with values of quantities of the same kind.

NOTE 1 A reference quantity value can be a true quantity value of a measurand, in which case it is unknown, or a conventional quantity value, in which case it is known

NOTE 2 A reference quantity value with associated measurement uncertainty is usually provided with reference to

(a) *a material, e.g., a certified reference material,*
(b) *a device, e.g., a stabilized laser,*
(c) *a reference measurement procedure,*
(d) *a comparison of measurement standards.*

Thus, considering the value of a measurement standard such as the "conventional value," the instrument's accuracy is related to its ability to present the measurement results as close as possible to the value of this standard.

Measurement accuracy is not a quantity and is not given a numerical quantity value. A measurement is said to be more accurate when it offers a smaller measurement error. [VIM—2.13 NOTE 1]

4.3.6.2 Measurement Precision

The definition in VIM—2.15 for measurement precision is:

Closeness of agreement between indications or measured quantity values obtained by replicate measurements on the same or similar objects under specified conditions.

NOTE 1 Measurement precision is usually expressed numerically by measures of imprecision, such as standard deviation, variance, or coefficient of variation under the specified conditions of measurement

NOTE 2 The 'specified conditions' can be, for example, repeatability conditions of measurement, intermediate precision conditions of measurement, or reproducibility conditions of measurement (see ISO 5725-1:1994)

NOTE 3 Measurement precision is used to define measurement repeatability, intermediate measurement precision, and measurement reproducibility

NOTE 4 Sometimes "measurement precision" is erroneously used to mean measurement accuracy

4.3.7 Measurement Precision × Measurement Accuracy

The following example is a "classic of metrology," but we consider it the simplest and fastest way to visually convey the concepts of precision and accuracy.

Consider four people (A, B, C, and D) who shoot ten times at the same distance as the target. The results of the shots are shown in Fig. 4.12.

4.3.8 Accuracy Class

The definition of accuracy class is as follows:

Class of measuring instruments or measuring systems that meet stated metrological requirements that are intended to keep measurement errors or instrumental measurement uncertainties within specified limits under specified operating conditions.

NOTE 1 An accuracy class is usually denoted by a number or symbol adopted by convention

NOTE 2 Accuracy class applies to material measures. [VIM—4.25]

Example 4.6

(a) According to Mercosur standard NM 215: 2000, a Class 1 standard block may have a variation in its length L (in mm) of $\pm$ (0.05 + 0.5 × 10^{-6} L) (m) /year).

(b) According to the recommendations of OIML, the standard masses used in the calibration of scales are classified in the accuracy classes E1, E2, F1, F2, M1, and M2. A mass of 100 mg, for example, presents, by accuracy class, the following maximum permissible errors (Table 4.9):

4.4 Repeatability and Reproducibility

The measurement result is intrinsically linked to these definitions. We can only compare results that meet the conditions of repeatability or reproducibility.

4.4.1 Repeatability Condition of Measurement

According to VIM—2.20, the repeatability condition of measurement is:

Condition of measurement, out of a set of conditions that includes the same measurement procedure, same operators, same measuring system, same operating conditions, and same location, and replicate measurements on the same or similar objects over a short period of time.

NOTE 1 A condition of measurement is a repeatability condition only with respect to a specified set of repeatability conditions

NOTE 2 In chemistry, the term "intra-serial precision condition of measurement" is sometimes used to designate this concept

Table 4.9 Maximum permissible error values for the mass of 100 mg

Accuracy class	Maximum error
Class E1	±0.005 mg
Class E2	±0.015 mg
Class F1	±0.05 mg
Class F2	±0.15 mg
Class M1	±0.5 mg
Class M2	±1.5 mg

4.4.2 Measurement Repeatability

The VIM—2.21 presents the following definition of repeatability of measurement: *"measurement precision under a set of repeatability conditions of measurement."*

4.4.3 Reproducibility Condition of Measurement

Again, the VIM—2.24 presents the following definition for the reproducibility condition of measurement:

> *Condition of measurement, out of a set of conditions that includes different locations, operators, measuring systems, and replicate measurements on the same or similar objects.*
> *NOTE 1 The different measuring systems may use different measurement procedures*
> *NOTE 2 A specification should give the conditions changed and unchanged, to the extent practical*

Measurements in reproducibility conditions are very common in exports, as it is not possible for the same operator in the same place, following the same measurement system, to accompany the product.

4.4.4 Measurement Reproducibility

According to VIM—2.25, measurement reproducibility is *"measurement precision under reproducibility conditions of measurement."*

In the case of exports, measurement reproducibility will verify the variability of measurements between places or countries. This variability must be within criteria previously established in the contract.

4.5 Proposed Exercises

4.5.1 According to the pressure gauge shown in Fig. 4.13, answer:

(a) What is the division of the scale?
(b) Which reading resolution would you adopt?
(c) How would you write the result of reading the gauge?

4.5.2 According to the thermometer shown in Fig. 4.14, answer:

(a) What is the division of the scale?
(b) Which reading resolution would you adopt?
(c) How would you write the result of reading the thermometer?